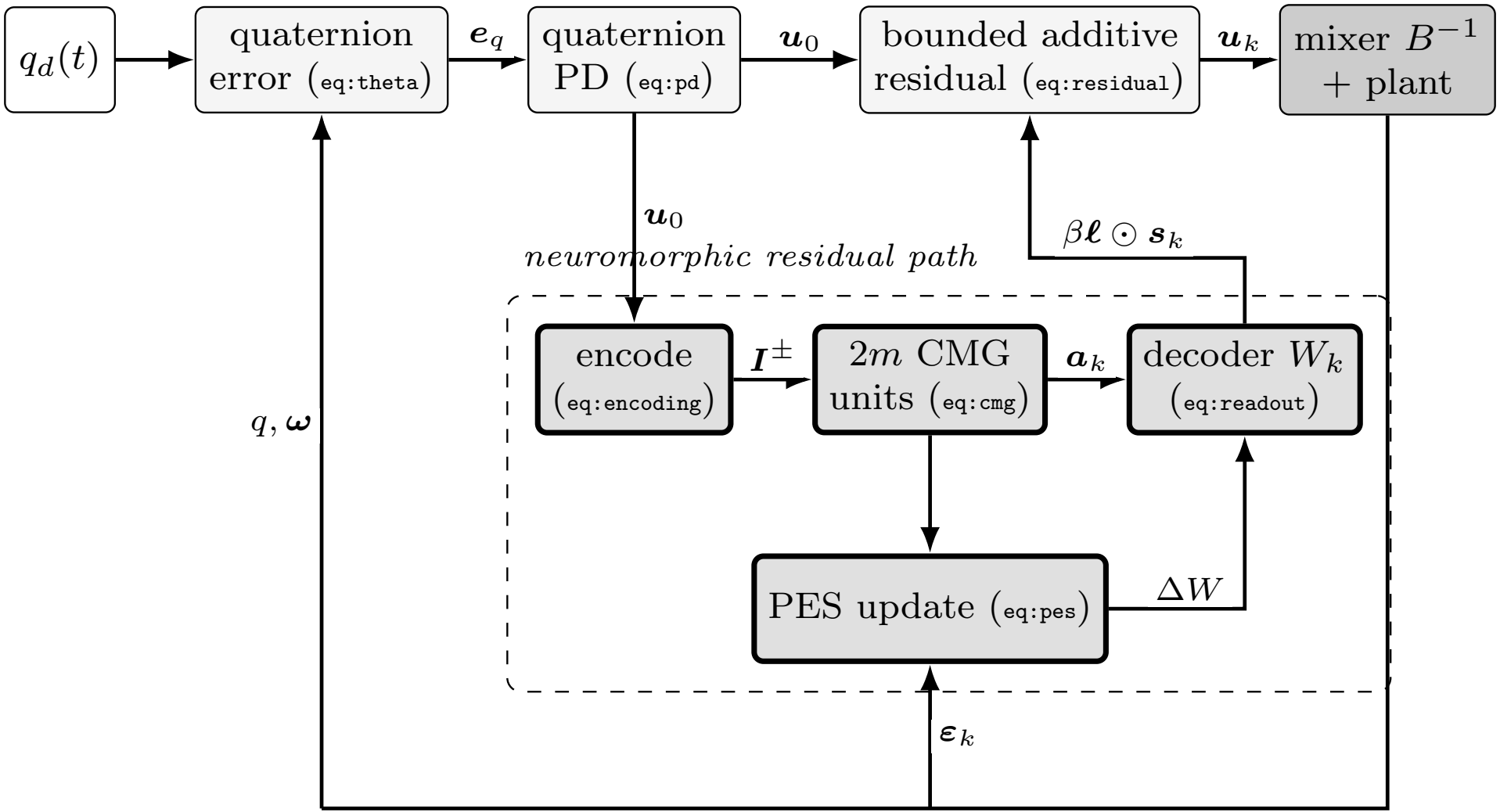
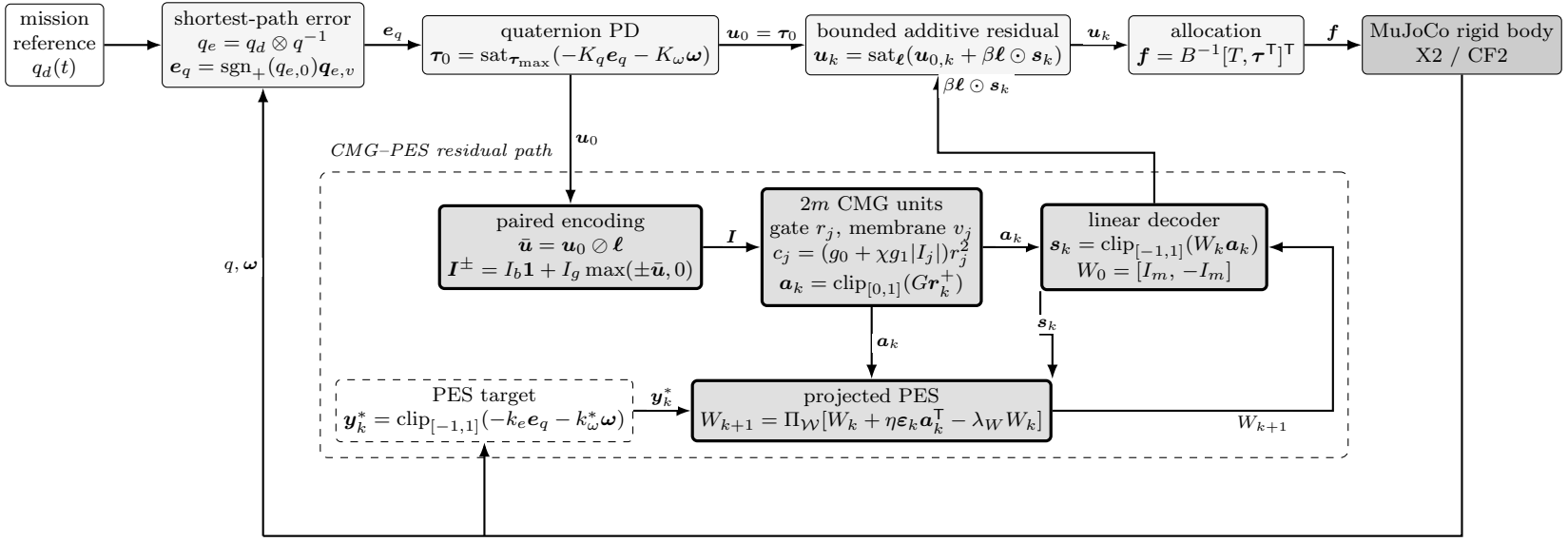
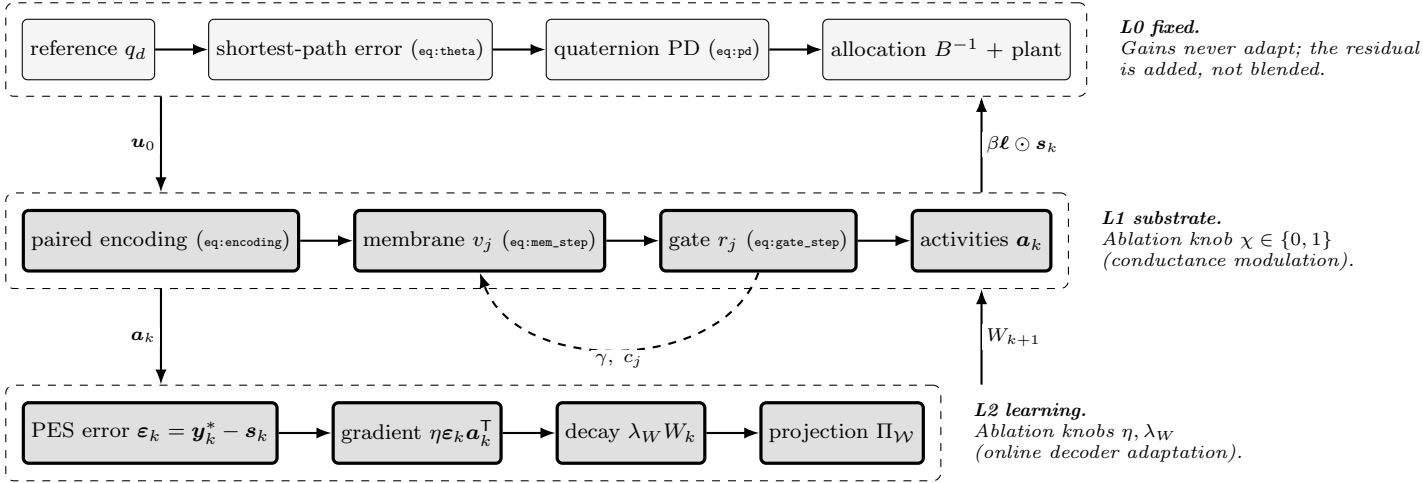

$q_d(t)$
quaternion error (eq:theta)
$e_q$
quaternion PD (eq:pd)
$u_0$
bounded additive residual (eq:residual)
$u_k$
mixer $B^{-1}$ + plant
$u_0$
neuromorphic residual path
$\beta \ell \odot s_k$
encode (eq:encoding)
$I^{\pm}$
$2m$ CMG units (eq:cmg)
$a_k$
decoder $W_k$ (eq:readout)
$q, \omega$
PES update (eq:pes)
$\Delta W$
$\varepsilon_k$


mission reference $q_d(t)$
shortest-path error $q_e = q_d \otimes q^{-1}$ $\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}$
$\boldsymbol{e}_q$
quaternion PD $\boldsymbol{\tau}_0 = \mathrm{sat}_{\boldsymbol{\tau}_{\max}}(-K_q\boldsymbol{e}_q - K_\omega\boldsymbol{\omega})$
$\boldsymbol{u}_0 = \boldsymbol{\tau}_0$
bounded additive residual $\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta\boldsymbol{\ell} \odot \boldsymbol{s}_k)$
$\boldsymbol{u}_k$
allocation $\boldsymbol{f} = B^{-1}[T, \boldsymbol{\tau}^\mathsf{T}]^\mathsf{T}$
$\boldsymbol{f}$
MuJoCo rigid body X2 / CF2
$\beta\boldsymbol{\ell} \odot \boldsymbol{s}_k$
$\boldsymbol{u}_0$
CMG–PES residual path
paired encoding $\bar{\boldsymbol{u}} = \boldsymbol{u}_0 \oslash \boldsymbol{\ell}$ $\boldsymbol{I}^\pm = I_b\mathbf{1} + I_g \max(\pm\bar{\boldsymbol{u}}, 0)$
$\boldsymbol{I}$
$2m$ CMG units gate $r_j$, membrane $v_j$ $c_j = (g_0 + \chi g_1|I_j|)r_j^2$ $\boldsymbol{a}_k = \mathrm{clip}_{[0,1]}(G\boldsymbol{r}_k^+)$
$\boldsymbol{a}_k$
linear decoder $\boldsymbol{s}_k = \mathrm{clip}_{[-1,1]}(W_k\boldsymbol{a}_k)$ $W_0 = [I_m, -I_m]$
$\boldsymbol{s}_k$
$q, \boldsymbol{\omega}$
PES target $\boldsymbol{y}_k^* = \mathrm{clip}_{[-1,1]}(-k_e\boldsymbol{e}_q - k_\omega^*\boldsymbol{\omega})$
$\boldsymbol{y}_k^*$
projected PES $W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta\boldsymbol{\varepsilon}_k\boldsymbol{a}_k^\mathsf{T} - \lambda_W W_k]$
$W_{k+1}$

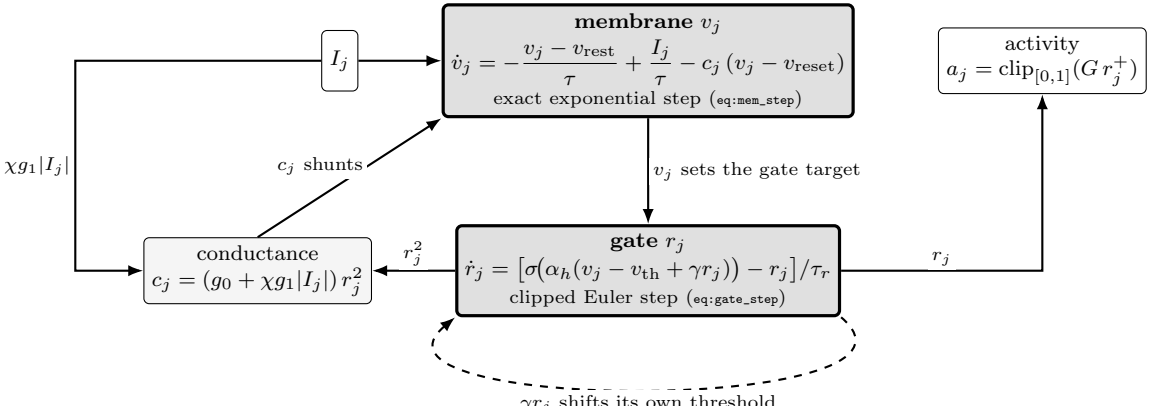



***L0 fixed.***
*Gains never adapt; the residual is added, not blended.*

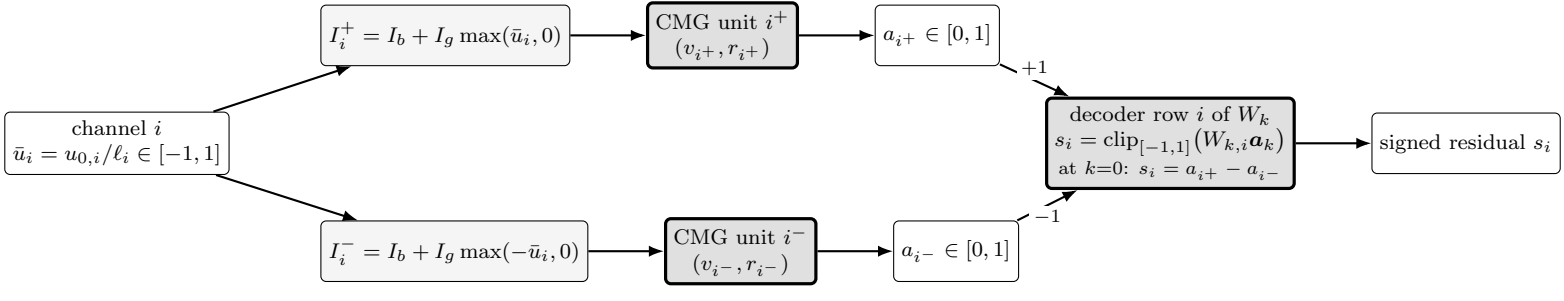
***L1 substrate.***
*Ablation knob* $\chi \in \{0, 1\}$
*(conductance modulation).*

***L2 learning.***
*Ablation knobs* $\eta, \lambda_W$
*(online decoder adaptation).*

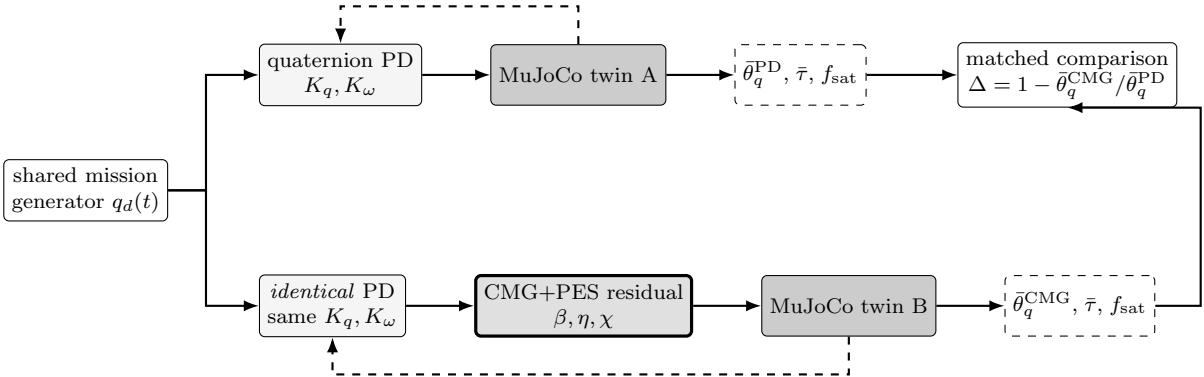
*Interface knob* $\beta \in [0, 1]$ *scales L1→L0 authority. Setting* $\beta$=0 *recovers the baseline exactly; setting* $\eta$=0 *freezes* $W_k$=$W_0$*; setting* $\chi$=0 *removes input-dependent conductance.*




*Two coupling paths make the unit* coupled *rather than a cascade: the gate shifts its own activation threshold ($\gamma$) and shunts the membrane through $c_j \propto r_j^2$. The dynamics are continuous — no spike and no discrete reset — and only $a_j$ leaves the unit. Setting $\chi$=0 removes the input-dependent part of the conductance and gives the ablated variant.*

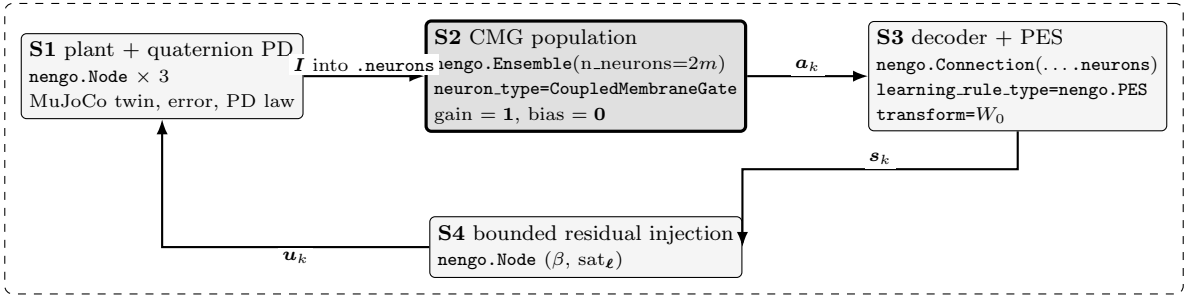



*Each of the $m$ command channels drives an antagonistic pair, so the population has $2m$ units. Activities are non-negative by construction; the sign is carried by the decoder. Initialising $W_0 = [I_m, -I_m]$ makes the untrained residual exactly the pair difference, so learning starts from a meaningful, not a random, readout.*

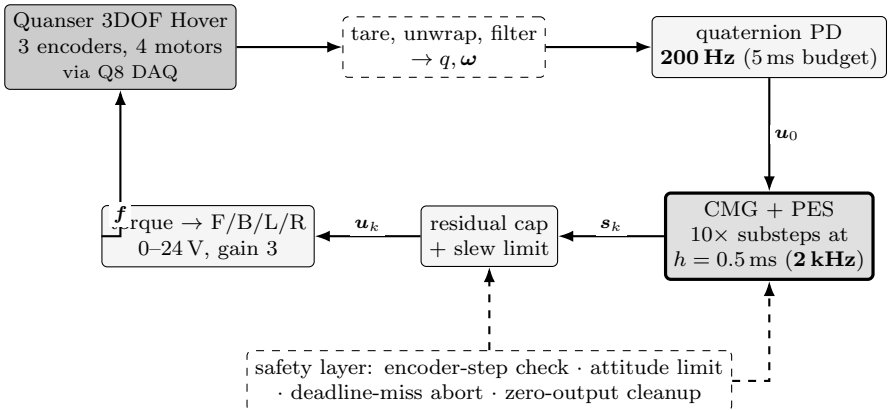



*Both twins share reference, duration, integration step, actuator scale, allocator, initial condition, safety limits and logging. The residual path is the only difference, so the reported improvement is attributable to it rather than to re-tuning.*

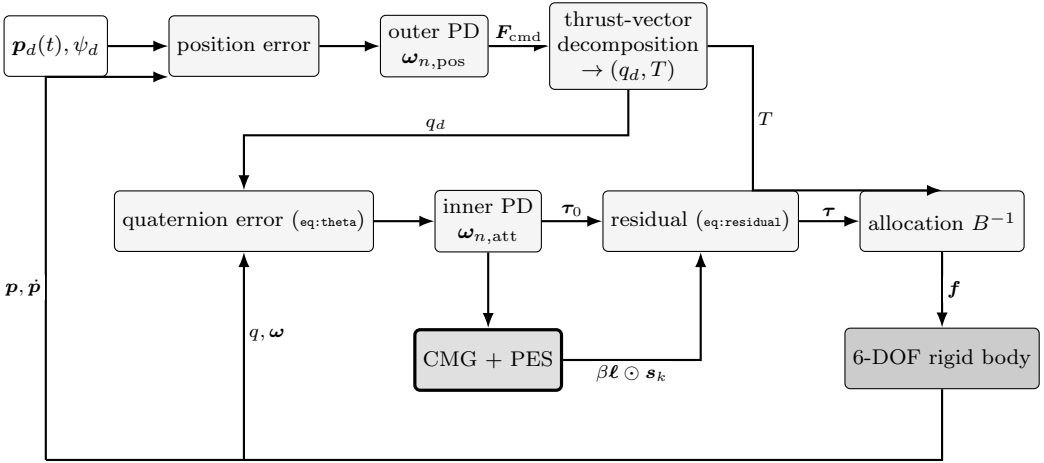
*built model: 1 Ensemble, 25 Nodes, 35 Connections, 1 PES rule*



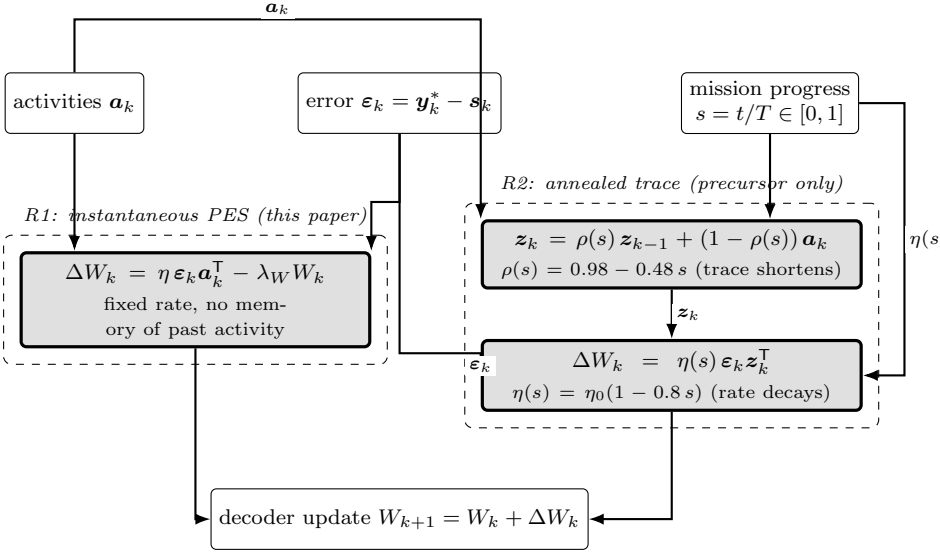

*The CMG population is the model's* only ***`Ensemble`****: in Nengo a custom* ***`NeuronType`*** *can exist only inside one. Everything else is a* **`Node`***, i.e. non-neural scaffolding. Current is injected at* **`.neurons`***, bypassing NEF encoders, and the PES learning rate is converted as* $\eta_{\text{nengo}} = \eta\,(2m)/h$ *so the two realizations coincide.*




*Hard real time: each 200 Hz hardware step contains ten 0.5 ms neural substeps, so the neural state advances at the same rate as in simulation. The controller is driven directly from Nengo; running the same model under the Nengo GUI produces interpreter-lock contention and deadline-miss aborts.*




*Time-scale separation is required, not optional: the validated ceiling is* $\boldsymbol{\omega}_{n,\mathrm{pos}} = [1.0, 1.0, 1.2]$ *against* $\boldsymbol{\omega}_{n,\mathrm{att}} = [4, 4, 3.5]$ *rad/s. Both* $q_d$ *and the measured* $q$ *must be expressed in the same quaternion convention before the error is formed.*




*Both rules share the same activities, the same error signal and the same projected decoder; they differ only in* what *multiplies the error and* how fast *it is applied. R2 explores early and consolidates late. Its outcome is mission dependent — beneficial in some missions, adverse in others — so it is reported separately rather than folded into the main results.*